

# D.I.Y Automatic irrigation system

## Preface:

This project aims to develop an irrigation system in two modes: manual, using an IR remote-controlled sprinkler, with more modes of irrigation controlled by a remote; automatic, the relay been activated by multiple sensors like, soil humidity, temperature and air humidity.

## Objectives:

Increase knowledge about Arduino, and microcontrollers.

Using multiple sensors, relay system, and water pump.

Learning the principles of automation of some equipment.

Improving knowledge about IR remote and the use of these technologies.

Improving knowledge regarding the principles of electronics.

Design of automatic-manual operation equipment.

## Components:

| Component                           | Role                                 |
|-------------------------------------|--------------------------------------|
| Arduino Mega 2560                   | Microcontroller                      |
| Breadboard                          | Use for connect components           |
| Soil humidity sensor                | Measure soil moisture                |
| Temperature+ humidity sensor(DHT11) | Measure air moisture and temperature |
| IR Remote + Receiver                | For manual control                   |
| Submerge water pump                 | Irrigation for plants                |
| 5V Relay module                     | Acvitate the water pump              |
| External 9V source                  | Power supply for the water pump      |

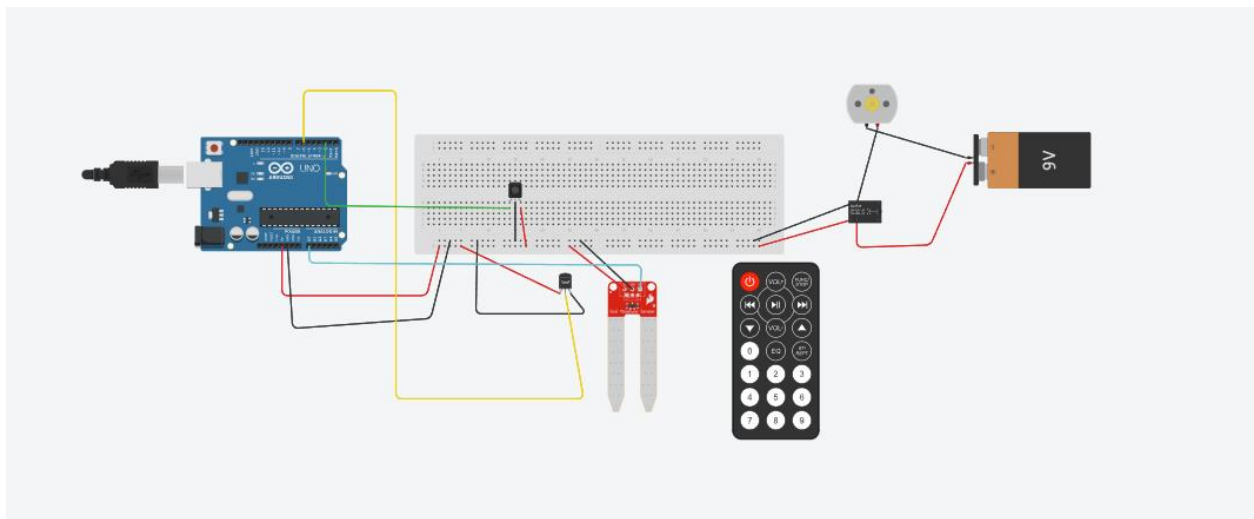
## Circuit diagram:

In thinkercard water pump doesn't exist, so I used in diagram a 5V DC motor, which has same principle of functioning.

Same for DHT11 sensor, I used a temperature sensor.

Because in thinkercard, relay module doesn't exist, I used one simple relay.

For any disagreements, I put on GitHub photos and video of this circuit, with all connections.



## Code:

\*First of all, for DHT11 and IRRemote system, we need to add some libraries:

DHT11: "DHT.h"

IRRemote: < IRremote.hpp>

\*Configure pin for every component connected on Arduino.

| Component                          | Pin number |
|------------------------------------|------------|
| IR Remote + Receiver               | 2          |
| Relay module                       | 5          |
| DHT11(Temperature+humidity sensor) | 6          |
| Soil humidity sensor               | A0         |

```
#define DHTPIN 6 // pin 6 for dht11 sensor
#define DHTTYPE DHT11
#define IR_RECEIVE_PIN 2 // pin 2 for ir remote

uint32_t lastCode = 0;
int releu = 5; // pin number 5 on microcontroller
int pinSenzor = A0; //analog pin A0 on microcontroller
long timer = 0; // timer used on manual mode

bool manual = true;
```

\*For manual mode,I used a bool type variable,which switch the system from automatic to manual mode and vice versa.

\*Setup all sensors,relay module,and water pump.

```
void setup() {
  Serial.begin(115200);
  IrReceiver.begin(IR_RECEIVE_PIN, ENABLE_LED_FEEDBACK);
  Serial.println("Pornit receptor IR...");

  pinMode(pinSenzor, INPUT);
  pinMode(releu, OUTPUT);
  digitalWrite(releu, HIGH);

  dht.begin();
}
```

pinSensor(Analogic pin for soil humidity sensor) is input,because this type of sensor read information,and sent to microcontroller.

\*Further,we test all sensors,storing the parameters in some variables,and to verify,print that values in serial port.

```
float temperatura = dht.readTemperature();// variable for temperature
float umiditate = dht.readHumidity();//variable for humidity of dht11
int valoarea = analogRead(pinSensor);//read the values of soil humidity sensor

Serial.print("Soil humidity: ");
Serial.println(valoarea);

Serial.print("Humiditate: ");
Serial.println(umiditate);
Serial.print("Temperature: ");
Serial.println(temperatura);

delay(1000);
```

\*For manual mode,we need store the pressing of one button in a variable,because when we press the remote,it transmit an code to our IRRemote from breadboard.

```
//manual mode
if (IrReceiver.decode()) {
  unsigned long cod = IrReceiver.decodedIRData.decodedRawData;
  Serial.print("Cod primit: ");
  //start-stop for relay
  if (cod == 0x184C) {
    digitalWrite(releu, HIGH);
  } else if (cod == 0x104C) {
    digitalWrite(releu, LOW);
  }
}
```

\*In manual mode,we have more modes of irrigation,first one is a simple start-stop system,acting by the same buton,which emits two semnals,like a bool variable.

\*Others mode are:in continue irrigation for 5 seconds,when the relay is active for 5 seconds,and take a short delay after.

Intermittent mode, where the relay is like a traffic light, active for a period equal to the stop period.

```
//the relay is activated for 5 seconds, after it's turn off
if (cod == 0x184D || cod == 0x104D) {
    digitalWrite(releu, LOW);
    delay(5000);
    digitalWrite(releu, HIGH);
    delay(2000);
}

//the relay is intermittent
if (cod == 0x1875 || cod == 0x1075) {
    digitalWrite(releu, LOW);
    delay(2000);
    digitalWrite(releu, HIGH);
    delay(2000);
}
```

\*For this modes, we need to press that buttons.

\*Automatic mode is activated after some times, after the manual mode doesn't work. This mode is the priority, and is activated every time we don't use the remote.

Here we setup some program of irrigation, in function of soil humidity, temperature, and air temperature.

```
//mode one, for high temperature, and dry soil and moderate humidity
if (temperatura > 25 && valoarea >= 500 && umiditate <= 40) {
    digitalWrite(releu, LOW);
    delay(10000);
    digitalWrite(releu, HIGH);
    delay(500);
}

//mode two
if (temperatura >= 20 && temperatura <= 30 && valoarea > 500 && umiditate >= 40 && umiditate < 60) {
    digitalWrite(releu, LOW);
    delay(5000);
    digitalWrite(releu, HIGH);
    delay(500);
}

//mode 3
if (temperatura >= 30 && valoarea >= 450 && valoarea <= 500 && umiditate <= 40) {
    digitalWrite(releu, LOW);
    delay(5000);
    digitalWrite(releu, HIGH);
    delay(500);
}
```

\*For an improvement of automatic mode,we ad an extra mode,which stop the irrigation,when the soil is too wet,the others factors are excluded.

```
//mode four,the soil is very wet,so we don't need extra factors
if (valoare < 350) {
    digitalWrite(releu, HIGH);
    delay(500);
}
```

## Results and tests:

After testing the program and circuits in different modes and situations,we had next problems and difficulties:

\*Between IRRemote and remote is a little delay,which can make manual mode havier.

\*In my weather conditions,I can't test the thirt mode of automatics.

## Conclusions and improvements:

The project demonstrated hardware-software integration in a simple embedeed project,very useful for future IRRemote project,automation projects,and learn to work with multiple sensors.

### Improvements:

\* Integration of manual mode and through an application.

\*Add a irrigation mode based on photoresistor(a resista which increase the resistance depending on the brightness).

\*Add a water sensor to prevent overflowing in pots.

\*Add a clock module to irrigate on different hours.

**Others:**

Full code + video of functionality: <https://github.com/EdwardM88/Diy-Automated-irrigation-system>

Linkedin: <https://www.linkedin.com/in/edward-merticariu-0535522b6/>